

Java Interview Questions - Topicwise Guide

Topic: Java Strings (20 Questions by Level)

Q1. What is the difference between `==` and `.equals()` for Strings?

`==` checks reference equality, `.equals()` checks content equality.

Example:

```
String a = new String("hello");
String b = new String("hello");
System.out.println(a == b); // false
System.out.println(a.equals(b)); // true
```

Q2. Why is String immutable in Java?

Reasons: Thread safety, security, caching, and class loading benefits.

Example:

```
String a = "java";
a.concat("8");
System.out.println(a); // java
```

Q3. What is the String pool?

A memory area for storing unique String literals.

Example:

```
String s1 = "Java";
String s2 = "Java";
System.out.println(s1 == s2); // true
```

Q4. How are StringBuilder and StringBuffer different from String?

String is immutable. StringBuilder is mutable and faster. StringBuffer is mutable and thread-safe.

Example:

```
StringBuilder sb = new StringBuilder("hello");
sb.append(" world");
```

Q5. How do you reverse a String in Java?

Use StringBuilder's reverse method.

Example:

```
String reversed = new StringBuilder("interview").reverse().toString();
```

Q6. What does the intern() method do?

Places a String in the pool or returns existing pooled String.

Example:

```
String a = new String("java").intern();
String b = "java";
```

Java Interview Questions - Topicwise Guide

```
System.out.println(a == b); // true
```

Q7. How does `substring()` work internally?

Java 6: shared char array. Java 7+: creates new char array.

Example:

```
String str = "HelloWorld";  
System.out.println(str.substring(5)); // World
```

Q8. What happens when using `String.valueOf(null)`?

Returns the String "null", does not throw exception.

Example:

```
System.out.println(String.valueOf(null)); // null
```

Q9. Difference between `replace()` and `replaceAll()`?

`replace()` uses character/String, `replaceAll()` uses regex.

Example:

```
"abc123".replaceAll("\\d", ""); // abc
```

Q10. How to remove whitespace from a String?

Use `trim()` or regex.

Example:

```
" Java ".trim(); // "Java"
```

Q11. Can you extend String class?

No. String is final and cannot be extended.

Example:

```
class MyString extends String {} // Compilation error
```

Q12. How is `hashCode()` of String calculated?

Using polynomial rolling hash function.

Example:

```
hash = 31 * hash + char
```

Q13. String concatenation in loop - performance concern?

Inefficient as it creates new objects. Use `StringBuilder` instead.

Example:

```
StringBuilder sb = new StringBuilder();
```

Q14. Can two identical Strings be stored in different memory?

Yes, if created using new operator vs literal.

Example:

Java Interview Questions - Topicwise Guide

```
String a = new String("abc");
```

```
String b = "abc";
```

Q15. How to handle sensitive String data securely?

Use char[] and overwrite it to avoid lingering in memory.

Q16. What is the output of 'he' + 'llo' == 'hello'?

true - compile-time constant folding applies.

Q17. Output of String s1 = "he"; s1 + "llo" == "hello"?

false - runtime operation creates a new object.

Q18. null + "abc" results in?

"nullabc" - null is converted to string.

Q19. How to find first non-repeating character in a String?

Use LinkedHashMap to track frequencies and maintain order.

Q20. Does String s1 = "a"; s1 + "b" == "ab"?

false - runtime concat not pooled.